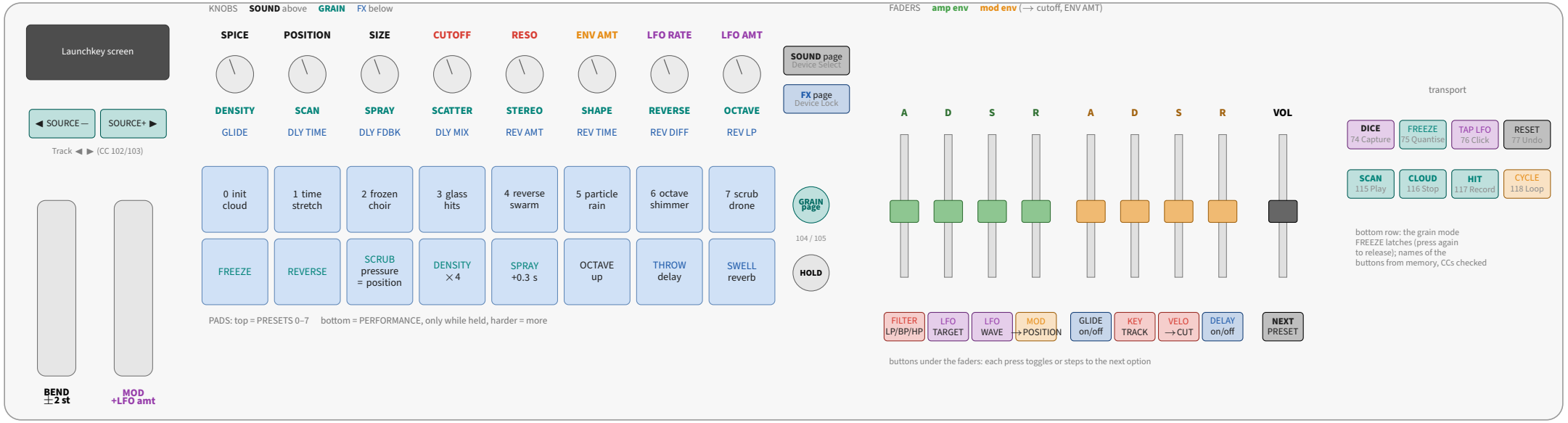
# grains

Axoloti Core + Launchkey 49 MK3 (standalone mode) · mono granular synth, the SD-card patch · `patches/grains/grains_v1.axp`






## Knobs: three pages, scaled pickup

| | |
|---|---|
| **SPICE** | chaos: each phrase gets its own position (±0.25), pitch (±0.5 st) and cutoff (±1.5 oct) offset, and all grains more spray (+0.1 s), scatter (+3 st), reversal (+0.3) and width (+0.5) |
| **POSITION** | where in the source, 0 – 1 (SCAN: where a phrase starts) |
| **SIZE** | grain length 5 ms – 1 s (exponential) |
| **knobs 4–8** | CUTOFF, RESO, ENV AMT, LFO RATE, LFO AMT: as freak |
| **DENSITY** | 0.5 – 200 grains/s; in HIT, grains per note, 1 – 20 |
| **SCAN** | SCAN speed: centre = still, right forward, left backward, $\frac{1}{16}$ – 4× |
| **SPRAY** | random start, ±0 – 0.5 s |
| **SCATTER** | random pitch, ±0 – 12 st |
| **STEREO** | random pan width, 0 – 1 |
| **SHAPE** | grain window: smooth (bell) → percussive (4 % attack) |
| **REVERSE** | chance a grain plays backwards |
| **OCTAVE** | chance a grain jumps an octave, up or down |
| FX page | as freak: GLIDE; delay (now stereo); reverb |

**Pages**: Device Select = SOUND, round button (upper, right of the pads) = GRAIN, Device Lock = FX. A knob left behind by a page change, preset or DICE moves the value proportionally until they meet: no jumps.

## Grain modes (Play / Stop / Record)

**SCAN**: grains at a steady rate from a play head that moves through the source at SCAN speed, from POSITION on each new phrase: time stretch, slow motion, reverse, still.

**CLOUD**: grains at random times (mean = DENSITY) around POSITION: textures.

**HIT**: each note fires DENSITY grains at once, then nothing: percussive. Short SHAPE and a plucky amp envelope suit it.

C4 (middle C) plays a source at its own pitch; keys transpose the grains (−48 to +36 st).

## Keys: mono, legato, hold

The newest key sounds; release it and the previous one comes back, without restarting (legato). A key with no other down starts a *phrase*: envelopes restart, SCAN jumps to POSITION, SPICE rolls. GLIDE as freak. **HOLD** (round button, lower) or the sustain pedal keeps the note after release; a new key then starts a new phrase.

## Envelopes, filter, LFO

As freak: faders 1–4 amp, 5–8 mod (1 ms – 10 s; S 0 – 1); CYCLE (Loop) loops the mod env. Mod env → cutoff (ENV AMT) and → position (+0.5) with MOD→POSITION. Grains keep coming while the amp envelope is open, release included. Filter: stereo 12 dB/oct SVF, LP/BP/HP. LFO TARGET: **position** (±0.5) → size (× $\frac{1}{4}$–4) → cutoff (±3 oct) → pitch (±2 st); WAVE and TAP as freak.

## Sources

**SOURCE** (Track ◀ / ▶) steps through what is loaded: 0 = built-in (a saw chord that brightens along its 1.5 s), 1–8 = the starter bank (page 2), then your files. Presets name a source; one that is not loaded plays the built-in one.

**Your own**: WAV files in `/grains` on the SD card, up to 16, loaded in name order (name them `09_xxx.wav`, at most 8 characters before `.wav`). 16, 24 or 32-bit or float; any rate; stereo is mixed to mono. Memory: 7.2 MB in all, 3 MB of it the starter bank, so about 45 s at 48 kHz for your own (the log shows what is left). Put them there with *Board → Enter Card Reader Mode*.

Loading runs in the background after power-up (the starter bank: 1.1 s): until a source is there, the built-in one plays.

## Pads

**Top** = presets 0–7. **Bottom** = performance, while held; strike velocity, then pressure, sets the amount:

| | |
|---|---|
| FREEZE | the position stops (SCAN head too) |
| REVERSE | grains backwards, SCAN runs backwards |
| SCRUB | pad pressure *is* the position |
| DENSITY / SPRAY | up to 4× / +0.3 s |
| OCTAVE | +12 st |
| THROW / SWELL | delay throw / reverb swell, as freak |

## Two patches, one board

**SD card in** at power-up: grains starts. **No card**: freak (in flash) starts.

**Install**: open `grains_v1.axp`, *Upload to SD Card as Startup*. That also copies the starter bank to `/grains` (and replaces an older startup patch on the card).

**If grains will not start**: power off, take the card out: freak starts. Then see page 2.

## Signal path and load

20 grains (pool) → stereo SVF → amp env × velocity → stereo delay → Elements reverb → VOL. When the pool is full, new grains wait: the load stays bounded. DSP 41 % idle, 75.5 % with all 20 grains; patch RAM 28 of 44 KB.

## Notes

Toggle states are not shown (no LEDs in standalone mode). With the patcher connected, its log reports the loaded files, pages, presets, modes, sources and knobs catching up.

## Starter bank: `/grains/01..08.wav`, 4 s each, made with freak's oscillators at C4

(each changes along its length, so POSITION matters)

| | |
|---|---|
| **1** | vowel FOF on C and G: a → e → i → o → u along the source |
| **2** | paraphonic chords, a new chord each second |
| **3** | harmonics, their centre rising (dark → bright) |
| **4** | bell, 8 strikes (one per 0.5 s), more inharmonic each time |
| **5** | particles, density and pitch spread rising |
| **6** | feedback FM, more modulation along it, a new ratio each second |
| **7** | plucked arpeggio C E G B D B G E, 8 notes a second |
| **8** | saw swarm C3 G3 C4, detune rising, filter opening |

## Presets: top pads = 0–7; NEXT PRESET (fader button 9) steps; RESET (Undo) = 0

(chosen from the parameter meanings, not yet tuned by ear)

| | |
|---|---|
| **0 init cloud** | CLOUD on source 1 (vowels), middle; 140 ms grains, 22/s, half width; LP 4.7 kHz; delay off. The power-on sound. |
| **1 time stretch** | SCAN source 2 (chords) from the start at half speed: every chord lasts twice as long, pitch unchanged. |
| **2 frozen choir** | SCAN still, source 1 at 0.43: 320 ms grains, 44/s, wide, some octaves; slow triangle LFO drifts the position; attack 0.33 s, release 3 s. Hold a note. |
| **3 glass hits** | HIT on source 4 (bells) at a strike: 11 percussive grains per note, wide, some octaves; decay 0.7 s. Play rhythms. |
| **4 reverse swarm** | CLOUD on source 8 (saw swarm): 87 % of the grains backwards, S&H LFO on grain size. |
| **5 particle rain** | CLOUD on source 5 (particles): 40 ms grains at 90/s, spray 0.25 s, ±2.7 st, full width, percussive; S&H LFO on cutoff. |
| **6 octave shimmer** | CLOUD on source 3 (harmonics): long grains, 80 % jump an octave; attack 0.7 s, release 3 s; big reverb. |
| **7 scrub drone** | SCAN source 7 (plucks) at 0.07×: a frozen arpeggio slowly turning; CYCLE on; glide. |

A preset sets the mode, the source, all three knob pages, faders 1–8 and the toggles; the volume stays. **DICE** (Capture MIDI) rolls the mode and the SOUND and GRAIN pages instead; source, envelopes, FX and volume stay.

## Sound design moves

**Time freeze**: SCAN with SCAN at the centre, or hold FREEZE; then sweep POSITION by hand or SCRUB with pressure. **Texture from anything**: CLOUD, SIZE 1/2, DENSITY 3/4, SPRAY 1/4, STEREO full. **Glitch**: HIT, SIZE 1/4, SHAPE full, SCATTER 1/2, play fast. **Moving drone**: LFO TARGET position, slow triangle, MOD→POSITION with CYCLE on. **Pitched clouds**: SCATTER 0, OCTAVE 1/2, the Launchkey's Scale mode.

## Where the values fall (knob turned . . . )

| | 1/4 | 1/2 | 3/4 | full |
|---|---|---|---|---|
| **SIZE** | 19 ms | 72 ms | 0.26 s | 1 s |
| **DENSITY** (grains/s) | 2.3 | 10 | 44 | 200 |
| **DENSITY** in HIT (grains/note) | 6 | 11 | 15 | 20 |
| **SCAN** (× real time) | −0.45 | still | +0.42 | +4 |
| **SPRAY** (±) | 32 ms | 0.13 s | 0.28 s | 0.5 s |
| **SCATTER** (±) | 0.8 st | 3 st | 6.7 st | 12 st |

SCAN: still from 61 to 67 (the centre); ×1 at about 5/6 of the travel (and −1 at 1/6). Envelopes, CUTOFF, RESO, ENV AMT, LFO, GLIDE, delay and VOL fall as on the freak card.

## Something seems wrong?

**No sound**: VOL up? In HIT, sound only comes on a key press. SIZE and DENSITY not both at the bottom? CUTOFF as on freak.

**Only the built-in sound**: no `/grains` folder, or the files are still loading or not WAV (the log says which). Upload the patch to the SD card again to copy the bank.

**Sound stuck on**: HOLD or the sustain pedal. Press HOLD (round button, lower).

**Nothing moves**: FREEZE latched (Quantise), or SCAN at its centre.

**grains does not start, or hangs**: power off, remove the card, power on: freak starts from flash. Fix the card with Card Reader Mode (rename `/start.bin` to skip grains).

**Lost**: RESET (Undo).

**Another patch** (on the SD card): hold both buttons left of the pads and tap a top pad: 1 freak, 2 grains, 3 groove, 4 soliton, 5 aeolian, 6 raw, 7 quartet.

## MIDI reference (beyond the freak card)

| | |
|---|---|
| GRAIN page / HOLD (round buttons) | ch 16 CC 104 / 105 |
| SCAN, CLOUD, HIT (Play, Stop, Record) | ch 16 CC 115, 116, 117 |
| FREEZE latch (Quantise) | ch 16 CC 75 |
| SOURCE − / + (track ◀/▶) | ch 16 CC 102 / 103 |
| sustain pedal (= HOLD while down) | ch 1 CC 64 |